JET RESIDUES AT INFINITY AND COLLISION MODULI OF CUBIC-FRAME KELLER MAPS

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Abstract. An admissible pair of one-variable polynomials

$$A(0) = 0, \quad A'(0) = 1, \qquad B(0) = -2, \quad B'(0) = -2A''(0)$$

defines a polynomial Keller map $G_{A,B} \colon \mathbb{A}^3 \to \mathbb{A}^3$ of generic degree three. We remove the squarefreeness hypothesis from the boundary-residue classification. If $\gcd(A, B) = 1$, put

$$Z_A = \operatorname{Spec} \mathbb{C}[c]/(A/c) \subset \mathbb{G}_m, \qquad \sigma_{A,B} = B|_{Z_A} \in H^0(Z_A, \mathcal{O}_{Z_A})^*.$$

The finite scheme Z_A records the unmarked sheets that disappear at infinity; when it is nonreduced, $\sigma_{A,B}$ records their full transverse jet. We prove that the decorated scheme $(Z_A, \sigma_{A,B})$, modulo scaling of the coordinate c, is a complete invariant under arbitrary polynomial left–right equivalence after every affine stabilization. Stable equivalence already implies ordinary left–right equivalence.

For $\deg A = N + 1$, the orbit set is parametrized by the geometric points of a natural weighted quotient

$$\mathfrak{B}_N = [\mathcal{U}_N / \mathbb{G}_m],$$

where $\mathcal{U}_N$ is the unit scheme in the rank-N algebra over $\operatorname{Hilb}^N(\mathbb{G}_m)$. An explicit affine presentation makes $\mathfrak{B}_N$ a smooth Deligne–Mumford boundary-data stack of dimension $2N - 1$. Its squarefree locus is the configuration space of labelled points; its collision locus replaces colliding labels by derivatives. In particular, a single unmarked point of multiplicity m carries an m-parameter family of pairwise stably inequivalent maps, all at generic degree three and ordinary degree $4m + 7$. The original modulus $q = \beta/\alpha^2$ is the first-order case: it is the residue of the forbidden global Tschirnhaus transformation.

1. The picture

The affine Jacobian condition removes local ramification. It does not remove the possibility that sheets of a finite cover leave the affine chart. For the maps in this paper, the inverse equation is

$$P_{a,b,c}(T) = A(c)T^3 + B(c)T^2 + bT - 2a. \tag{1}$$

The zeros of A are the parameters at which one root moves to infinity. The zero at $c = 0$ is marked and retained in the affine source. The finite scheme

$$Z_A = V(A/c) \subset \mathbb{G}_m$$

records all other infinity sheets, including their collision multiplicities.

The relevant rational function is

$$\tau_{A,B}(c) = -\frac{B(c)}{3A(c)}. \tag{2}$$

On the normalization of the discriminant, the triple-root curve is the graph $t = \tau_{A,B}(c)$. A fiberwise Tschirnhaus translation $T \mapsto T + \phi(c)$ changes τ by subtraction of the regular function ϕ. Consequently, the polar part of τ along the unmarked divisor Z_A cannot be removed by a polynomial change of root. Multiplication by A identifies this polar part with

$$\sigma_{A,B} = B \bmod (A/c) \in H^0(Z_A, \mathcal{O}_{Z_A}).$$

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If Z_A is reduced, this is a list of values. If several infinity sheets collide, it is a list of jets.

The main result is a boundary Torelli theorem:

$$\boxed{\text{the stable left–right class is reconstructed from the polar data at infinity.}}$$

It has two consequences that are easy to miss if one only studies the squarefree locus.

(1) Collisions do not destroy moduli. Values at distinct points converge to derivatives on a nonreduced point.
(2) For every N, the entire length-N collision space is smooth at the level of boundary data, even though the associated discriminant geometry changes visibly.

2. Universal cubic frames

Put

$$c = 2x - 3x^2y - x^3z, \qquad t = y + \frac{1}{x}, \qquad r = \frac{2}{x}. \tag{3}$$

For $A, B \in \mathbb{C}[c]$, let

$$h(T, c) = A(c)T^3 + B(c)T^2$$

and define

$$b = r - h_T(t, c), \qquad 2a = h(t, c) + tb. \tag{4}$$

Definition 2.1. The pair (A, B) is *admissible* if

$$A(0) = 0, \quad A'(0) = 1, \qquad B(0) = -2, \quad B'(0) = -2A''(0). \tag{5}$$

We write

$$G_{A,B} = (a, b, c) \colon \mathbb{A}^3_{x,y,z} \longrightarrow \mathbb{A}^3_{a,b,c}$$

for the map defined by (3)–(4).

Proposition 2.2 (Universal Keller identity)**.** *For every admissible pair, the expressions a, b, c are polynomials and*

$$\det DG_{A,B} = -2.$$

If $A \neq 0$, the generic degree is three.

Proof. Write

$$A(c) = c + c^2U(c), \qquad u_0 = U(0) = \frac{A''(0)}{2}, \qquad B(c) = -2 - 4u_0c + c^2V(c).$$

Set $\xi = 1 + xy$ and $d = 2 - 3xy - x^2z$, so that $c = xd$ and $t = \xi/x$. Direct substitution gives

$$b = \frac{2 + 4\xi - 3d\xi^2}{x} - 3d^2U(c)\xi^2 + 8u_0d\xi - 2xd^2V(c)\xi,$$

$$2a = \frac{2\xi - 2d\xi^3 + 2\xi^2}{x^2} + \frac{-2d^2U(c)\xi^3 + 4u_0d\xi^2}{x} - d^2V(c)\xi^2.$$

The universal numerators in the first fractions are divisible by x and x^2, respectively. Writing $U(c) = u_0 + cW(c)$, the remaining numerator is divisible by x because $2 - d\xi$ is. Hence a, b are polynomial.

On $x \neq 0$,

$$\det \frac{\partial(t, r, c)}{\partial(x, y, z)} = -2x, \qquad \det \frac{\partial(a, b, c)}{\partial(t, r, c)} = \frac{r}{2}.$$

Since $r = 2/x$, the product is -2, and polynomiality extends the identity across $x = 0$.

For a target point, every finite simple root of (1) reconstructs

$$x = \frac{2}{P'(t)}, \qquad y = t - \frac{P'(t)}{2}, \qquad z = \frac{2x - 3x^2y - c}{x^3}. \tag{6}$$

For general c, the cubic has three finite simple roots. $\square$

Lemma 2.3 (The retained infinity section)**.** *The restriction of $G_{A,B}$ to $V(x)$ is an isomorphism onto $V(c)$. More precisely, if*

$$u_0 = \frac{A''(0)}{2}, \qquad u_1 = \frac{A'''(0)}{6}, \qquad v_0 = \frac{B''(0)}{2},$$

then

$$G_{A,B}(0, y, z) = \Big(z + 4y^2 + 2u_0y - 2v_0 - 8u_1,\ y + 4u_0,\ 0\Big). \tag{7}$$

Thus the infinity root over $c = 0$ is retained in the affine source.

3. Finite completion and polar data

Assume for the rest of the paper that

$$\gcd(A, B) = 1. \tag{8}$$

No squarefreeness assumption is imposed. Put

$$Q_A(c) = \frac{A(c)}{c}, \qquad Z_A = \operatorname{Spec} \mathbb{C}[c]/(Q_A).$$

Since $Q_A(0) = 1$, this is a finite subscheme of $\mathbb{G}_m$.

Homogenize the inverse cubic:

$$\overline{P}_{a,b,c}(U, V) = A(c)U^3 + B(c)U^2V + bUV^2 - 2aV^3. \tag{9}$$

Let

$$\overline{X}_{A,B} = V(\overline{P}) \subset \mathbb{A}^3_{a,b,c} \times \mathbb{P}^1_{[U:V]}.$$

Projection to the target is finite of degree three: it is projective, and (8) prevents a whole projective line from occurring in a fiber.

Near infinity, use $s = V/U$. The equation is

$$A(c) + B(c)s + bs^2 - 2as^3 = 0. \tag{10}$$

At every point of Z_A, the derivative in the root direction is the unit $B|_{Z_A}$. Hence the infinity branch is simple as a root of the cubic, even when its base scheme is nonreduced.

Proposition 3.1 (Affine source as a marked-root chart)**.** *The source $\mathbb{A}^3$ is isomorphic to the open subset of $\overline{X}_{A,B}$ obtained by removing the ramification divisor and the infinity branch over Z_A, while retaining the infinity branch over $c = 0$.*

Proof. On the finite-root chart, the isomorphism and its inverse are given by (6). The root is simple because $P'(t) = 2/x$. The retained infinity branch is identified with $V(x)$ by theorem 2.3. The only remaining simple infinity points lie over Z_A, and no source point with $x = 0$ can map there because $c = 0$ on $V(x)$. $\square$

The Tschirnhaus meaning of the invariant is now immediate. The triple-root curve on the normalized discriminant will be

$$t = \tau_{A,B}(c) = -\frac{B(c)}{3A(c)}.$$

A root translation by $\phi(c)$ changes τ by $-\phi$. The polar part along Z_A is therefore invariant under polynomial translations.

Definition 3.2 (Jet residue)**.** The *jet residue at infinity* is

$$\sigma_{A,B} = B \bmod Q_A \in H^0(Z_A, \mathcal{O}_{Z_A})^*. \tag{11}$$

It is a unit precisely because $\gcd(A, B) = 1$.

For fixed A, the admissible polynomials form an affine space

$$B_0 + c^2\mathbb{C}[c], \qquad B_0 = -2 - 2A''(0)c.$$

A polynomial root translation must satisfy $\phi(0) = 0$, and changes B by $3A\phi$. Hence

$$\frac{c^2\mathbb{C}[c]}{cA(c)\mathbb{C}[c]} \simeq \frac{\mathbb{C}[c]}{(Q_A)} = H^0(Z_A, \mathcal{O}_{Z_A}). \tag{12}$$

Thus the terminology "jet residue" is literal: a reduced point contributes a value, while a point of multiplicity m contributes m Taylor coefficients.

4. The intrinsic marked boundary

The cubic discriminant is

$$\Delta_{A,B} = B^2b^2 - 4Ab^3 + 8B^3a - 108A^2a^2 - 36ABab. \tag{13}$$

Let

$$D_{A,B} = V(\Delta_{A,B}).$$

For each closed point r in the support of Z_A, put $P_r = V(c - r)$.

Proposition 4.1 (Nonproperness boundary)**.** *The reduced nonproperness set is*

$$S_{G_{A,B}} = D_{A,B} \cup \bigcup_{r \in |Z_A|} P_r. \tag{14}$$

The discriminant component is irreducible and singular; every other component is an affine plane.

Proof. By theorem 3.1, the map is the restriction of a finite map and the deleted boundary consists of the ramification divisor and the unmarked infinity branch. Their images are $D_{A,B}$ and the planes P_r, respectively. The image of the deleted boundary of a finite completion is exactly the nonproperness set.

As a quadratic in a, $\Delta_{A,B}$ has discriminant

$$\mathrm{Disc}_a(\Delta_{A,B}) = 64(B^2 - 3Ab)^3.$$

The factor $B^2 - 3Ab$ is nonconstant linear in b, hence is not a square in $\mathbb{C}(b, c)$. The quadratic is primitive because $\gcd(A, B) = 1$. Gauss's lemma gives irreducibility. Singularity follows from the normalization below. □

Proposition 4.2 (Universal generic monodromy)**.** *For every coprime admissible pair, the inverse cubic is irreducible over $\mathbb{C}(a, b, c)$, and its Galois closure has group S_3. In particular, the degree-three function-field extension defined by $G_{A,B}$ has no nontrivial automorphism over the target field.*

Proof. Put $K_0 = \mathbb{C}(b, c)$ and

$$f(T) = A(c)T^3 + B(c)T^2 + bT.$$

The rational function $f\colon \mathbb{P}^1 \to \mathbb{P}^1$ has degree three, so $[K_0(T) : K_0(f(T))] = 3$. After identifying the transcendental element $2a$ with $f(T)$, this says that $f(X) - 2a$ is irreducible over $K_0(a)$. Its discriminant is $\Delta_{A,B}$, which is irreducible and therefore not a square in $\mathbb{C}(a, b, c)$. An irreducible cubic with nonsquare discriminant has Galois group S_3. The corresponding index-three subgroup is an S_2, whose normalizer in S_3 is itself; hence the cubic extension has trivial automorphism group over the base. □

This proposition emphasizes that the moduli below are invisible to the coarsest generic-cover data: every member has the same degree and the same full symmetric monodromy. What varies is the embedding of that cover at infinity.

A repeated finite root t gives

$$b = -3A(c)t^2 - 2B(c)t, \qquad a = -A(c)t^3 - \frac{1}{2}B(c)t^2. \tag{15}$$

Let

$$\nu_{A,B} \colon \mathbb{A}^2_{c,t} \longrightarrow D_{A,B}$$

be this map and put

$$H_{A,B}(c,t) = 3A(c)t + B(c). \tag{16}$$

Proposition 4.3 (Normalization and markings)**.** *The map $\nu_{A,B}$ is the normalization of $D_{A,B}$. Moreover,*

$$\nu_{A,B}^{-1}(\operatorname{Sing} D_{A,B}) = L_{A,B} := V(H_{A,B}). \tag{17}$$

For every $r \in |Z_A|$, the inverse image of $D_{A,B} \cap P_r$ is the line

$$M_r = V(c - r),$$

and $M_r \to D_{A,B} \cap P_r$ is an isomorphism.

Proof. On the parametrization, t satisfies

$$3At^2 + 2Bt + b = 0, \qquad Bt^2 + 2bt - 6a = 0.$$

A Bezout identity between A and B gives a monic quadratic equation for t, so $\nu_{A,B}$ is finite. The identities

$$B^2 - 3Ab = H_{A,B}^2, \qquad -18Aa - Bb = 2tH_{A,B}^2 \tag{18}$$

recover t rationally away from $H_{A,B} = 0$. Hence the map is birational, and $\mathbb{A}^2$ is normal.

The pulled-back gradient is

$$\begin{aligned}
(\Delta_{A,B})_a \circ \nu_{A,B} &= 8H_{A,B}^3,\\
(\Delta_{A,B})_b \circ \nu_{A,B} &= -4tH_{A,B}^3,\\
(\Delta_{A,B})_c \circ \nu_{A,B} &= -4t^2(A'(c)t + B'(c))H_{A,B}^3.
\end{aligned}$$

Thus the singular preimage is exactly $L_{A,B}$.

At a support point r, one has $A(r) = 0$ and $B(r) \neq 0$, so

$$a = -\frac{B(r)}{2}t^2, \qquad b = -2B(r)t.$$

The inverse is $t = -b/(2B(r))$. $\square$

The reduced nonproperness divisor therefore recovers the marked pair

$$\left(\mathbb{A}^2_{c,t}, L_{A,B}, \bigcup_{r \in |Z_A|} M_r\right). \tag{19}$$

The multiplicities in Z_A are not inserted as external markings. They will be recovered from the pole orders of $L_{A,B}$.

5. Stable jet-residue Torelli

Two polynomial maps are *stably left–right equivalent* if, after adjoining the same number of identity coordinates, they become equivalent under independent polynomial automorphisms of source and target.

Theorem 5.1 (Stable jet-residue classification)**.** *Let (A, B) and $(\widetilde{A}, \widetilde{B})$ be admissible coprime pairs. The following are equivalent.*

(i) *$G_{A,B}$ and $G_{\widetilde{A},\widetilde{B}}$ are stably left–right equivalent.*
(ii) *They are ordinarily left–right equivalent.*
(iii) *There exists $u \in \mathbb{C}^*$ such that*

$$\widetilde{A}(uc) = uA(c), \qquad \widetilde{B}(uc) - B(c) \in cA(c)\mathbb{C}[c]. \tag{20}$$

(iv) *Multiplication by u identifies the decorated finite schemes*

$$(Z_A, \sigma_{A,B}) \simeq (Z_{\widetilde{A}}, \sigma_{\widetilde{A},\widetilde{B}}). \tag{21}$$

Equivalently,

$$\widetilde{Q}(uc) = Q(c), \qquad \widetilde{B}(uc) \equiv B(c) \pmod{Q(c)}, \quad Q = A/c.$$

In particular, stabilization creates no new equivalences, and the theorem remains valid when Z_A is nonreduced.

Proof. The implication (ii)⇒(i) is immediate. Assume (i). The nonproperness divisor is stable under products with affine space. Its unique singular component must map to the unique singular component, and the union of its plane components maps to the corresponding union. Normalization commutes with adjoining polynomial variables, so the target equivalence lifts to an automorphism of a cylinder over $\mathbb{A}^2_{c,t}$ preserving $L_{A,B}$ and the reduced union of vertical lines.

If Z_A is empty, then $A = c$, and the conclusion follows directly from the gauge calculation below. Assume $Z_A \neq \varnothing$. Let C, T be the pullbacks of the target normalization coordinates and put

$$R_A = \operatorname{rad}(Q_A), \qquad R_{\widetilde{A}} = \operatorname{rad}(Q_{\widetilde{A}}).$$

Since the cylinder coordinate ring is a UFD, preservation of the reduced vertical divisor gives

$$R_{\widetilde{A}}(C) = \lambda R_A(c) \tag{22}$$

for some $\lambda \in \mathbb{C}^*$. Thus C is algebraic over $\mathbb{C}(c)$. The field $\mathbb{C}(c)$ is algebraically closed in the purely transcendental extension $\mathbb{C}(c, t, u_1, \dots, u_m)$, hence $C \in \mathbb{C}(c)$. Since C is polynomial, $C \in \mathbb{C}[c]$. Applying the same argument to the inverse automorphism gives $c \in \mathbb{C}[C]$, and therefore

$$C = uc + v, \qquad u \in \mathbb{C}^*, \ v \in \mathbb{C}. \tag{23}$$

The polynomial $H_{A,B} = 3At + B$ is primitive and linear in t, hence irreducible. Preservation of the singular preimage gives

$$3\widetilde{A}(C)T + \widetilde{B}(C) = \kappa(3A(c)t + B(c)). \tag{24}$$

Because T is polynomial and $C \in \mathbb{C}[c]$, the polynomial $\widetilde{A}(C)$ divides $A(c)$. Applying the same reasoning to the inverse automorphism gives the reverse divisibility after substituting (23); hence

$$\widetilde{A}(C) = \mu A(c) \tag{25}$$

for some $\mu \in \mathbb{C}^*$. Equation (22) maps every unmarked support point to an unmarked support point. The only remaining root of each polynomial is the marked simple root 0, so $C(0) = 0$. Thus $v = 0$. Differentiating (25) at the origin gives $\mu = u$, proving the first equation in (20).

Equation (24) now has the form

$$3uA(c)T + \widetilde{B}(uc) = \kappa\big(3A(c)t + B(c)\big).$$

It follows that

$$T = \frac{\kappa}{u}t + h(c).$$

At $c = 0$, the constant terms $B(0) = \widetilde{B}(0) = -2$ force $\kappa = 1$. Hence

$$\widetilde{B}(uc) - B(c) = -3uA(c)h(c). \tag{26}$$

The two admissibility conditions and the twice-differentiated identity $\widetilde{A}(uc) = uA(c)$ show that the left side has a double zero at 0. Since A has a simple zero there, $h(0) = 0$. This proves (iii).

Condition (iii) immediately identifies the decorated schemes. Conversely, (iv) gives $\widetilde{Q}(uc) = Q(c)$ and makes $D(c) = \widetilde{B}(uc) - B(c)$ divisible by $Q(c)$. Admissibility makes D divisible by c^2. Since $Q(0) = 1$, the ideals (c) and (Q) are comaximal, so D is divisible by $c^2Q = cA$. Thus (iii) and (iv) are equivalent.

It remains to construct an ordinary equivalence from (iii). Diagonal source and target scalings reduce to the case $\widetilde{A} = A$. Write

$$\widetilde{B} - B = 3A\phi, \qquad \phi \in c\mathbb{C}[c].$$

The source automorphism

$$\Theta_\phi(x, y, z) = \left(x,\ y + \phi(c),\ z - 3\frac{\phi(c)}{x}\right) \tag{27}$$

is polynomial because $\phi(c)/x = (c/x)(\phi(c)/c)$; it preserves c and sends t to $t + \phi(c)$. Put

$$L_\phi = 3A\phi^2 + 2B\phi, \qquad K_\phi = A\phi^3 + B\phi^2,$$

and define

$$\Xi_\phi(a, b, c) = \left(a - \frac{\phi(c)}{2}b - \frac{K_\phi(c)}{2},\ b + L_\phi(c),\ c\right). \tag{28}$$

Expanding $A(T + \phi)^3 + B(T + \phi)^2$ gives

$$G_{A,B+3A\phi} = \Xi_\phi \circ G_{A,B} \circ \Theta_\phi.$$

This proves (ii). □

Corollary 5.2 (Polar-part formulation)**.** *For fixed A, the complete stable invariant is the polar part of*

$$\tau_{A,B} = -\frac{B}{3A}$$

along Z_A, modulo regular functions vanishing at the marked point $c = 0$. Equivalently, it is the unit $\sigma_{A,B} \in H^0(Z_A, \mathcal{O}_{Z_A})^*$.

6. The collision space

Fix $N \geq 1$. Write

$$Q(c) = 1 + q_1c + \cdots + q_Nc^N, \qquad q_N \neq 0, \tag{29}$$

and

$$R(c) = r_0 + r_1c + \cdots + r_{N-1}c^{N-1}. \tag{30}$$

Set

$$A(c) = cQ(c), \qquad B(c) = -2 - 4q_1c + c^2R(c). \tag{31}$$

Every pair is admissible. Conversely, every gauge class with $\deg A = N + 1$ has a unique representative of this form: divide the coefficient of c^2 modulo Q.

Let

$$\mathcal{U}_N = \{(q_1, \dots, q_N, r_0, \dots, r_{N-1}) : q_N \operatorname{Res}_c(Q, B) \neq 0\} \subset \mathbb{A}^{2N}. \tag{32}$$

It is a smooth affine open subset. The unit condition is exactly $\operatorname{Res}(Q, B) \neq 0$.

The multiplicative group acts with weights

$$\lambda \cdot q_i = \lambda^i q_i, \qquad \lambda \cdot r_j = \lambda^{j+2} r_j. \tag{33}$$

Indeed, this is induced by

$$Q(c) \longmapsto Q(\lambda c), \qquad B(c) \longmapsto B(\lambda c),$$

which rescales the support of the infinity scheme.

Theorem 6.1 (Boundary-data quotient). *The geometric points of*

$$\mathfrak{B}_N = [\mathcal{U}_N / \mathbb{G}_m] \tag{34}$$

are in canonical bijection with stable left–right classes of coprime admissible cubic-frame maps with $\deg A = N + 1$.

The stack $\mathfrak{B}_N$ *is smooth Deligne–Mumford of dimension* $2N - 1$*. Its squarefree open substack is the quotient of labelled configuration data; its discriminant locus is the collision completion in which labels become jets.*

Proof. The first assertion is theorem 5.1 together with the unique normal form (31). The parameter space is smooth of dimension $2N$. Since $q_N \neq 0$, a stabilizer of the weighted action is contained in μ_N, hence is finite and reduced. The quotient is therefore a smooth Deligne–Mumford stack of dimension $2N - 1$.

The scheme defined by Q is the universal length-N subscheme of $\mathbb{G}_m$ in the standard presentation

$$\operatorname{Hilb}^N(\mathbb{G}_m) \simeq \operatorname{Spec} \mathbb{C}[q_1, \dots, q_N, q_N^{-1}].$$

The classes of B modulo Q form the rank-N algebra over this base, and the resultant condition selects its units. When Q is squarefree, a unit is a list of nonzero values. When roots collide, the same unit is a Taylor jet on the resulting Artin scheme. ☐

Remark 6.2 (What the stack does and does not claim). The stack $\mathfrak{B}_N$ is a canonical stack of decorated boundary data, and its geometric points classify stable map orbits. The theorem does not assert that its stabilizer groups equal the full automorphism groups of the polynomial maps; additional automorphisms acting trivially on the recovered boundary data have not been classified.

Proposition 6.3 (Tangent complex of the collision stack). *At a decorated finite scheme* (Z, σ)*, the tangent complex of* $\mathfrak{B}_N$ *is*

$$\left[\operatorname{Lie}(\mathbb{G}_m) \longrightarrow H^0(Z, N_{Z/\mathbb{G}_m}) \oplus H^0(Z, \mathcal{O}_Z)\right], \tag{35}$$

with the Lie algebra in degree -1*. Both summands on the right have dimension* N*. Thus the expected tangent dimension is* $2N - 1$*, at reduced configurations and at collision points alike.*

Proof. The Hilbert scheme of points on the smooth curve $\mathbb{G}_m$ has tangent space $H^0(Z, N_{Z/\mathbb{G}_m})$. The unit scheme is open in the rank-N algebra bundle, so its vertical tangent is $H^0(Z, \mathcal{O}_Z)$. Passing to the quotient by scaling gives the two-term quotient-stack tangent complex. ☐

The first summand deforms the support scheme, including its collision structure; the second deforms the transverse jet. At a multiple point these two directions remain independent, which is the infinitesimal reason the collision locus is smooth rather than a singular boundary of the moduli problem.

The squarefree locus has the familiar description

$$\frac{\left\{(r_i, v_i)_{i=1}^N \in (\mathbb{G}_m \times \mathbb{G}_m)^N : r_i \neq r_j\right\}}{\mathbb{G}_m \times S_N},$$

of dimension $2N-1$. The stack $\mathfrak{B}_N$ adds the diagonals without losing dimension or smoothness.

Proposition 6.4 (Collision turns values into derivatives)**.** *Let two support points be r and $r+h$, and prescribe labels*

$$v, \qquad v+hw.$$

As $h \to 0$, the decorated reduced scheme converges in $\mathfrak{B}_2$ to the double point

$$Z = \operatorname{Spec}\mathbb{C}[\varepsilon]/(\varepsilon^2), \qquad \varepsilon = c-r,$$

with unit section

$$\sigma = v + w\varepsilon.$$

Thus the divided difference of the two labels becomes the first jet at the collision point.

7. Higher-contact families

A particularly transparent family is obtained by concentrating the entire length at one point. For $m \geq 1$, put

$$A_m(c) = c(1-c)^m \tag{36}$$

and, for $\mathbf{s} = (s_0, \dots, s_{m-1}) \in \mathbb{C}^m$, put

$$B_{m,\mathbf{s}}(c) = -2 + 4mc + c^2 \sum_{j=0}^{m-1} s_j c^j. \tag{37}$$

The pair is admissible, and it is coprime precisely when

$$B_{m,\mathbf{s}}(1) = 4m - 2 + \sum_{j=0}^{m-1} s_j \neq 0. \tag{38}$$

Theorem 7.1 (Arbitrarily long jet moduli at one boundary point)**.** *For fixed m, the maps*

$$G_{A_m, B_{m,\mathbf{s}}}, \qquad \mathbf{s} \in \mathbb{C}^m \setminus \left\{\sum s_j = 2 - 4m\right\},$$

are pairwise stably left–right inequivalent. They all have

$$\det DG = -2, \qquad \textit{generic degree } 3,$$

and component degrees

$$(4m+7,\ 4m+6,\ 4). \tag{39}$$

Their complete invariant is the length-m jet

$$B_{m,\mathbf{s}}(1+\varepsilon) \bmod \varepsilon^m \in \left(\mathbb{C}[\varepsilon]/(\varepsilon^m)\right)^*.$$

Proof. Here $Z_{A_m} = m \cdot \{1\}$. Any scaling preserving this finite scheme fixes its support and is therefore the identity. The polynomials $\sum_{j=0}^{m-1} s_j c^j$ are the unique representatives modulo $(1-c)^m$, so distinct parameters give distinct jet residues.

Since $\deg A_m = m+1$ and $\deg B_{m,\mathbf{s}} \leq m+1$, the highest terms in the first two components come from $-A(c)t^3$ and $-3A(c)t^2$, respectively. Their degrees are $4(m+1)+3$ and $4(m+1)+2$; the third component has degree four. $\square$

Example 7.2 (The first genuine jet modulus)**.** For $m = 2$, take

$$A(c) = c(1-c)^2, \qquad B_{s,t}(c) = -2 + 8c + sc^2 + tc^3,$$

with $6 + s + t \neq 0$. These form a two-parameter family of pairwise stably inequivalent maps of component degrees $(15, 14, 4)$.

Writing $\varepsilon = c - 1$, one has

$$B_{s,t}(1) = 6 + s + t, \qquad B'_{s,t}(1) = 8 + 2s + 3t,$$

and

$$-\frac{B_{s,t}}{3A} = -\frac{6+s+t}{3\varepsilon^2} - \frac{2+s+2t}{3\varepsilon} + O(1). \tag{40}$$

Thus the two stable moduli are literally the double-pole and simple-pole coefficients of the forbidden Tschirnhaus gauge.

8. The boundary of the unit locus

The unit condition excludes ramified infinity sections. Its failure has a simple algebraic signature. Let $r \neq 0$, and set

$$m = \operatorname{ord}_r A, \qquad n = \operatorname{ord}_r B.$$

If $n = 0$, the infinity section is simple and the plane P_r is a separate nonproperness component. If $n > 0$, the infinity section is ramified and the plane is absorbed into the discriminant.

Proposition 8.1 (Discriminant content at a ramified infinity section)**.** *If $m \geq 1$ and $n \geq 1$, then the largest power of $c - r$ dividing the universal discriminant $\Delta_{A,B}$ is*

$$(c - r)^{\min(m,2n)}. \tag{41}$$

Proof. The five terms in (13) have $(c - r)$-orders

$$2n, \quad m, \quad 3n, \quad 2m, \quad m + n.$$

Their coefficients involve distinct monomials in a, b, so the lowest order cannot cancel. The minimum is $\min(m, 2n)$. ☐

For a simple zero of A, any vanishing label contributes the plane once. The exceptional quadratic value $q = -2$ is the case $m = 1$, $n = 2$: the boundary derivative vanishes, the deleted infinity section becomes ramified, and the discriminant acquires its plane factor.

Let $\mathcal{V}_N$ denote the full rank-N algebra over $\mathrm{Hilb}^N(\mathbb{G}_m)$, with $\mathcal{U}_N \subset \mathcal{V}_N$ its unit open. The complement $\mathcal{V}_N \setminus \mathcal{U}_N$ is therefore a natural candidate boundary for the orbit problem. Its local strata are indexed by the support multiplicities of Z and the vanishing orders of σ. The stable classification on this nonunit boundary is not proved here: the primitive discriminant normalization contracts positive-dimensional loci, as already occurs at the exceptional quadratic parameter.

9. Perspective and next problems

The preceding results suggest three related interpretations.

A Mittag–Leffler obstruction. Fiberwise, the quadratic term of a cubic can always be removed. Globally, the required Tschirnhaus translation is meromorphic in c. The jet residue is its principal part at the deleted divisor. The problem is thus a very elementary Mittag–Leffler problem with the additional demand that the gauge be polynomial and vanish at the marked point.

Spectral boundary data. The finite scheme Z_A is the spectral divisor of the sheet at infinity, and $\sigma_{A,B}$ is its first transverse coefficient. Collisions replace separate eigenvalue-like labels by nilpotent jets. This is why the Hilbert scheme, rather than only the configuration space, is the natural parameter space.

Marked cancellation. The stable theorem is a cancellation statement for the marked normalized boundary. Affine cylinders can forget substantial geometry, but the pair

$$\left(\mathbb{A}^2, L_{A,B}, \bigcup M_r\right)$$

retains the entire finite scheme and its jet residue. Stabilization adds rational directions but cannot manufacture the missing principal parts.

Question 9.1 (Nonunit compactification)**.** Does the quotient of the full algebra bundle $\mathcal{V}_N$ by scaling, with an appropriate modification along the resultant divisor, represent a natural compactification of stable cubic-frame orbits? The content formula (41) predicts its first boundary stratification.

Question 9.2 (Automorphism groups)**.** Can one prove that every stable automorphism of a generic cubic-frame map is generated, on the recovered boundary, by a scaling symmetry of $(Z_A, \sigma_{A,B})$? This would upgrade $\mathfrak{B}_N$ from a boundary-data stack with correct geometric points to the actual moduli stack of the restricted family.

Proposition 9.3 (The first boundary invariant in every degree)**.** *Let*

$$P(T) = A_n(c)T^n + A_{n-1}(c)T^{n-1} + \cdots + A_0(c)$$

be any one-parameter family of degree-n polynomials. Under a root translation $T \mapsto T + \phi(c)$,

$$A_{n-1} \longmapsto A_{n-1} + nA_n\phi.$$

In the local coordinate $s = 1/T$ at infinity, the equation begins

$$A_n(c) + A_{n-1}(c)s + O(s^2) = 0.$$

Hence the restriction

$$A_{n-1}|_{V(A_n)}$$

is simultaneously the first transverse coefficient of the infinity sheet and the first obstruction to a global polynomial Tschirnhaus normalization. For a nonreduced zero scheme it is again jet-valued.

This universal observation suggests that the cubic theory isolates the first layer of a hierarchy. In higher degree, once the T^{n-1}-coefficient is normalized, lower coefficients should contribute successive boundary data, while the partition stratification of the discriminant supplies the intrinsic markings needed for completeness.

Question 9.4 (Higher marked-root degree)**.** For inverse polynomials of degree greater than three, the normalized discriminant carries strata indexed by root partitions. Are the principal parts along deleted infinity schemes, together with these partition strata, complete stable invariants?

Moral.

> The maps differ neither in affine ramification, which is absent, nor in generic degree, which is fixed. They differ in the polar jet of the root normalization required at infinity. Values are the first moduli; collisions turn them into derivatives; polynomial coordinate changes cannot supply the missing poles; and affine stabilization cannot erase the marked boundary.

Verification boundary. The accompanying exact script checks the universal discriminant and normalization identities, the frame Jacobian, the marked section, the Tschirnhaus gauge, the weighted action, the double-point family, and sample content orders. The stable Torelli argument is geometric and is not replaced by symbolic verification. Independent specialist review remains appropriate before submission.